

Executive Analytics Programme

DATA SCIENCE IN MILITARY AVIATION

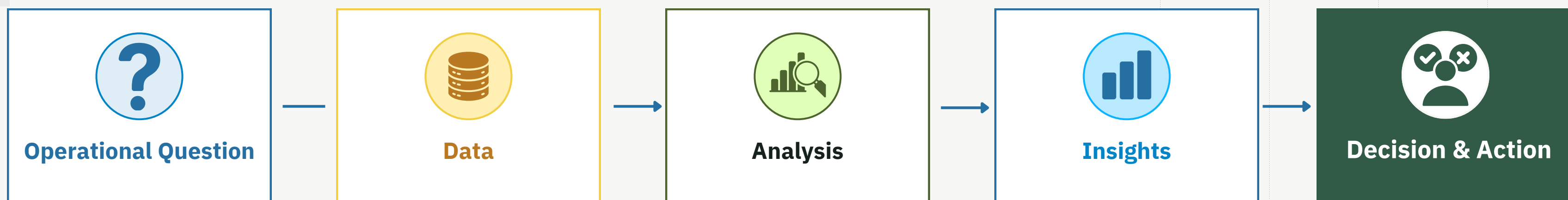
Understanding Why Data Matters in Modern Aviation Management

by
Henotes Data Dossier

From Analysis to Operational Decision

Yesterday's data answered operational questions.
Today the findings are brought together.

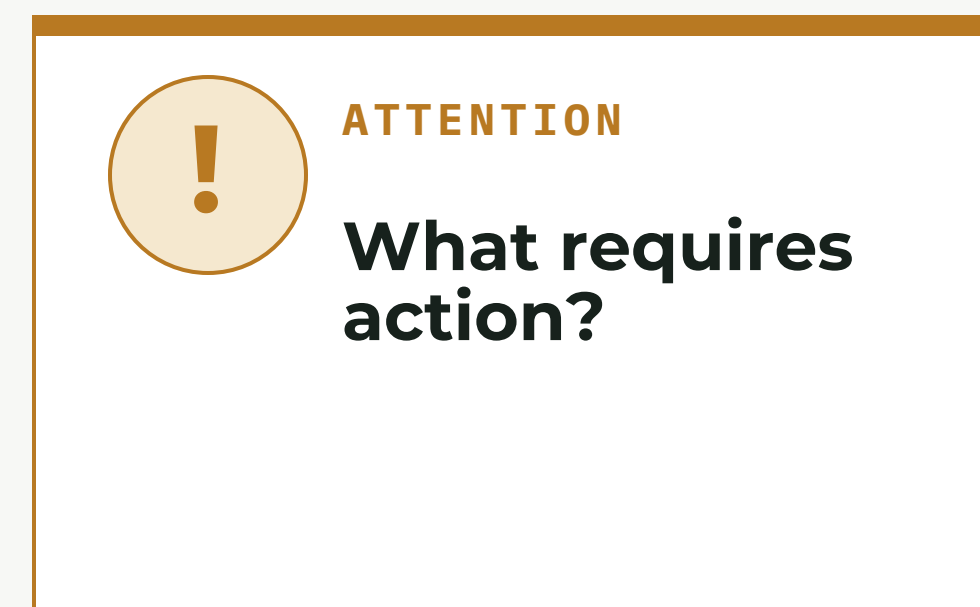
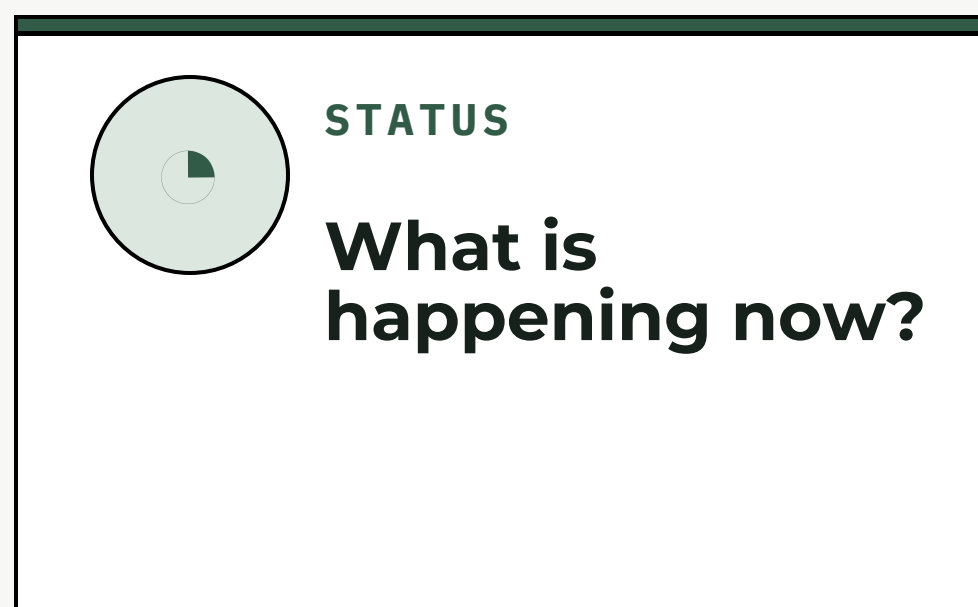
DESTINATION



**“The purpose of analytics is not to
produce charts.
It is to improve decisions.”**

What Is an Operational Dashboard?

“An operational dashboard is a visual summary of important information designed to help leaders monitor performance and identify issues requiring attention.”



A dashboard should reduce the time between seeing a problem and responding to it.

Anatomy of the Operational Logistics Dashboard

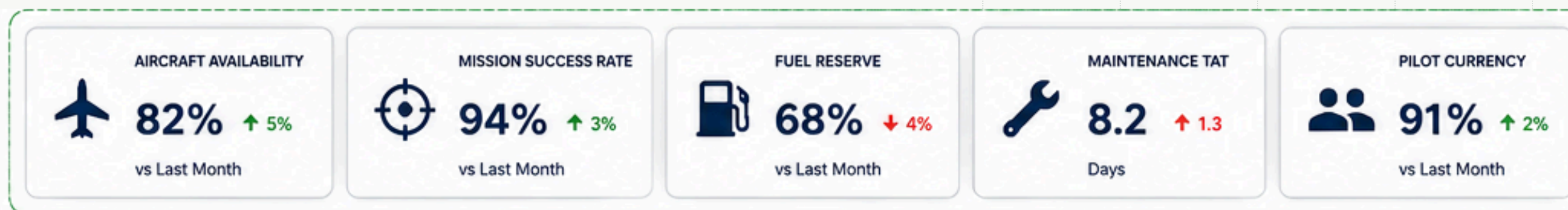
Title

DASHBOARD TITLE

Brief description of what the dashboard shows

The Dashboard title is used to name the dashboard. It is the name that gives an overall idea to what type of dashboard a commander is looking at.

KPI



The KPI section provides a quick snapshot of key performance indicators, allowing commanders to understand the current situation at a glance.

SLICERS/FILTERS

 **FILTERS / SLICERS**

DATE RANGE

01 May 2025 - 31 May 2025 

SQUADRON

All 

AIRCRAFT TYPE

All 

BASE

All 

STATUS

All 

Allow users to focus the dashboard on specific information such as a particular date, squadron, aircraft type, base, or status.

Chart Section

CHARTS & INSIGHTS

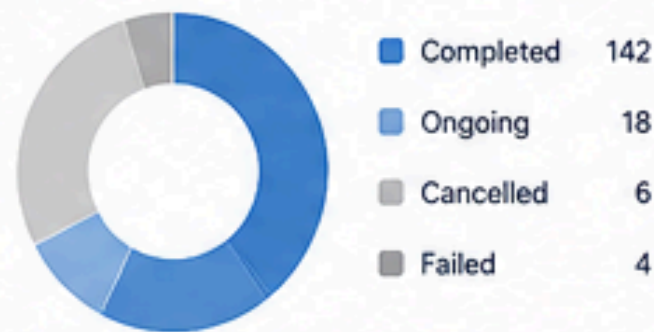
AIRCRAFT AVAILABILITY OVER TIME



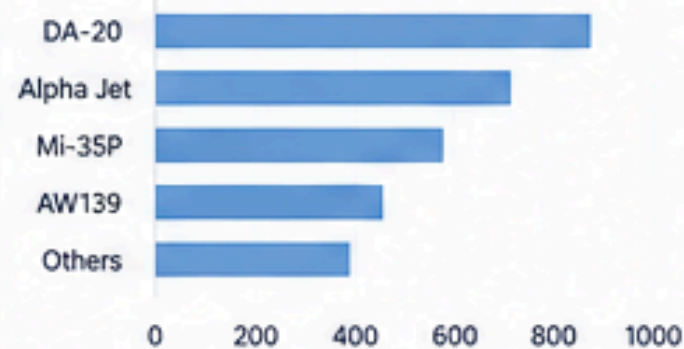
MISSION SUCCESS RATE OVER TIME



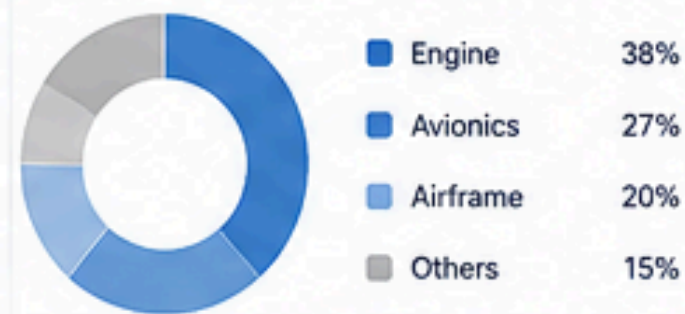
MISSION STATUS (MTD)



FLYING HOURS BY AIRCRAFT TYPE (MTD)



MAINTENANCE DOWNTIME BY CATEGORY

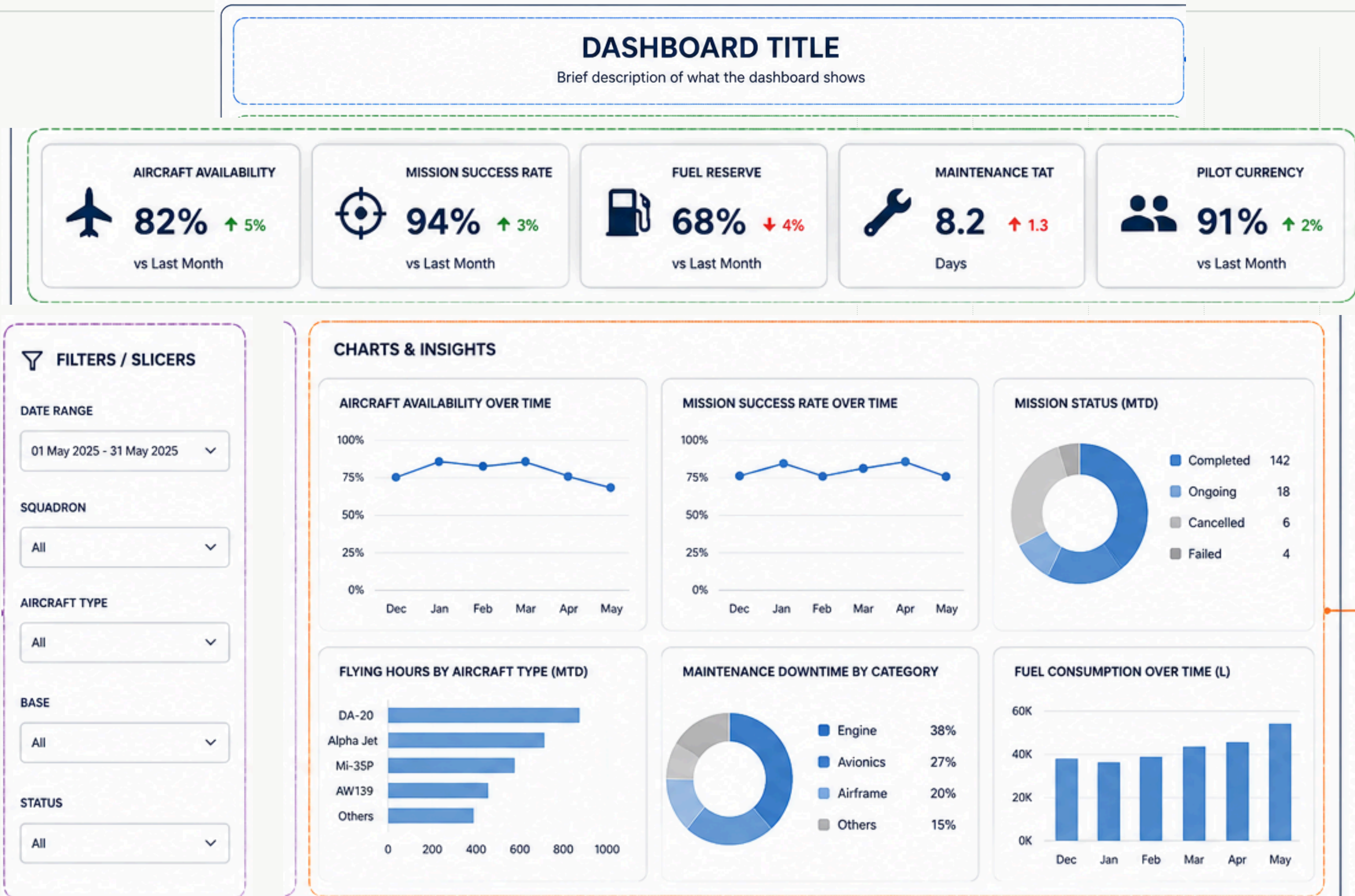


FUEL CONSUMPTION OVER TIME (L)

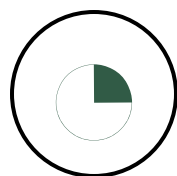


It uses charts and visual summaries to reveal trends, comparisons, patterns, relationships and exceptions, helping commanders quickly understand what is happening and identify areas that may require attention or action.

Anatomy of the Operational Logistics Dashboard

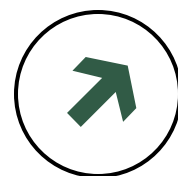


Reading the Dashboard: What Should We Look For?



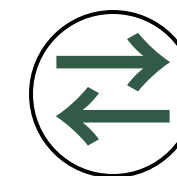
STATUS

What is happening now?



TREND

Is performance improving or declining?



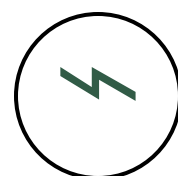
COMPARISON

Where is performance better or worse?



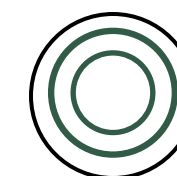
EXCEPTION

What stands out?



RELATIONSHIP

What indicators may be connected?



IMPLICATION

Why does this matter operationally?

Don't ask only 'What does the chart say?' Ask 'What does it mean?'

From KPI to Insight



KPI → Trend → Cause → Implication → Action



When Indicators Tell a Bigger Story

AIRCRAFT AVAILABILITY

54% ↓

MISSION SUCCESS

87% ↓

MAINTENANCE TURNAROUND

8.2 days ↑

FUEL CONSUMPTION

15% ↑

PILOT CURRENCY

91%

Higher flying activity may be increasing aircraft utilisation and maintenance demand, contributing to longer turnaround times and declining availability.

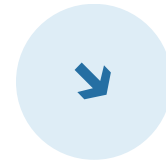
Analytics connects the numbers to the operational story.

From Insight to Command Recommendation



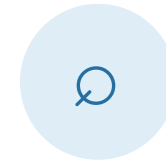
PROBLEM

Aircraft availability has declined.



EVIDENCE

Availability fell from 72% to 54% over six months.



INSIGHT

Three recurring defect categories account for 63% of downtime.



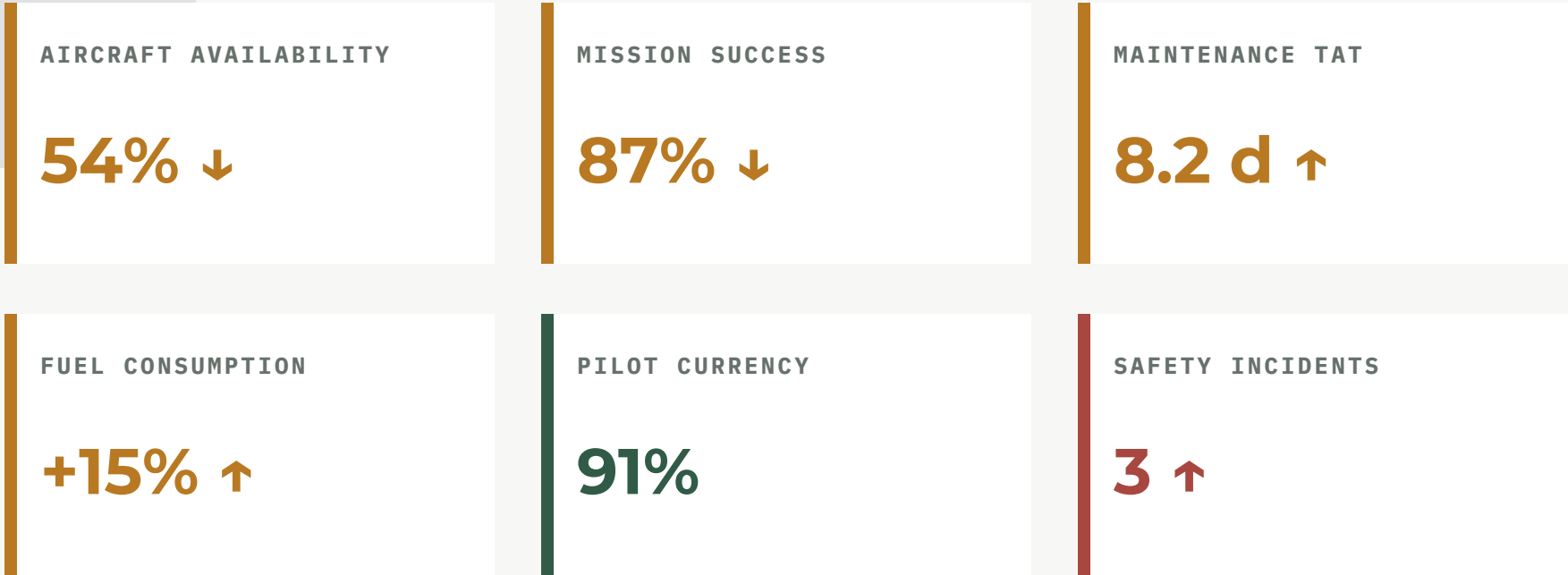
RECOMMENDATION

Prioritise the identified components and maintenance interventions, then monitor availability over the next 30 days.

A good recommendation is traceable to evidence.



What Would You Recommend to the Commander?



COMMANDER ASKS

What should I do?

PRIORITISE

What should receive immediate attention?

INVESTIGATE

What requires deeper analysis?

ACT

What recommendation would you make?

Data does not make the decision for the Commander. It gives the Commander better evidence with which to make the decision.

What Would You Recommend to the Commander?



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MAINTENANCE TAT

8.2 d ↑

FUEL CONSUMPTION

+15% ↑

PILOT CURRENCY

91%

SAFETY INCIDENTS

3 ↑

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LIVE DASHBOARD



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